```
'Ke'Shawn Stafford
'Project 12
'05/03/2025

' {$STAMP BS2}
' {$PBASIC 2.5}

DIRS = %11111111110
OUTS = %00000000000

LS    PIN 0
servo PIN 15
lcd   PIN 14

time     VAR Word
counter  VAR Byte
offset   VAR Byte
offset = 200

Main:
  SEROUT lcd, 84, [22, 12]
  PAUSE 250
  SEROUT lcd, 84, [208]

DO
  COUNT LS, 50, time
  PAUSE 10
  time = time - offset

  SELECT time

  CASE < 550
    SEROUT lcd, 84, [128, "Light Level: 1", 218, 220]
    FOR counter = 0 TO 2 : PULSOUT servo, 1230 : PAUSE 20 : NEXT
    OUTS = %11111111110

  CASE 600 TO 725
    SEROUT lcd, 84, [128, "Light Level: 2", 218, 222]
    FOR counter = 0 TO 2 : PULSOUT servo, 1140 : PAUSE 20 : NEXT
    OUTS = %11111111100

  CASE 750 TO 875
    SEROUT lcd, 84, [128, "Light Level: 3", 218, 223]
    FOR counter = 0 TO 2 : PULSOUT servo, 1030 : PAUSE 20 : NEXT
    OUTS = %11111111000

  CASE 900 TO 1000
    SEROUT lcd, 84, [128, "Light Level: 4", 218, 225]
    FOR counter = 0 TO 4 : PULSOUT servo, 920 : NEXT
    OUTS = %11111110000

  CASE 1025 TO 1150
    SEROUT lcd, 84, [128, "Light Level: 5", 218, 227]
    FOR counter = 0 TO 4 : PULSOUT servo, 810 : NEXT
    OUTS = %11111100000

  CASE 1175 TO 1300
    SEROUT lcd, 84, [128, "Light Level: 6", 218, 228]
    FOR counter = 0 TO 2 : PULSOUT servo, 700 : PAUSE 20 : NEXT
    OUTS = %11111000000

  CASE 1325 TO 1450
    SEROUT lcd, 84, [128, "Light Level: 7", 218, 230]
    FOR counter = 0 TO 2 : PULSOUT servo, 590 : PAUSE 20 : NEXT
    OUTS = %11110000000
```

```
  CASE 1475 TO 1600
    SEROUT lcd, 84, [128, "Light Level: 8", 219, 220]
    FOR counter = 0 TO 2 : PULSOUT servo, 480 : PAUSE 20 : NEXT
    OUTS = %11100000000

  CASE 1625 TO 1750
    SEROUT lcd, 84, [128, "Light Level: 9", 219, 222]
    FOR counter = 0 TO 2 : PULSOUT servo, 370 : PAUSE 20 : NEXT
    OUTS = %11000000000

  CASE > 1775
    SEROUT lcd, 84, [128, "Light Level: 10", 219, 223]
    FOR counter = 0 TO 2 : PULSOUT servo, 260 : PAUSE 20 : NEXT
    OUTS = %10000000000

  ENDSELECT

LOOP
```